



INFERENTIAL STATISTICS

FAST MCQ SHORTCUT TRICKS – BY THUNDERSTUDY

Population (N) = Entire Group | Sample (n) = Selected Few. Inferential statistics uses the Sample to guess about the Population.

✓ 1. Population vs Sample & Parameter vs Statistic

Memory Trick: **P** → Population → **P**arameter | **S** → Sample → **S**tatistic

Keywords: Whole/All = Population | Few/Selected = Sample

✓ 2. Hypothesis Testing - Magic Keywords

Statement Type	Identify By (Sign)
Null Hypothesis (H_0)	Equality ($=, \leq, \geq$)
Alternative (H_1)	Not equal ($\neq, <, >$)

Example: Mean = 50 is H_0 | Mean $\neq$ 50 is H_1

✓ 3. Errors - The Courtroom Analogy

- **Type I Error:** Rejecting H_0 when it's actually TRUE (Innocent punished).
- **Type II Error:** Accepting H_0 when it's actually FALSE (Criminal freed).

Ultra Shortcut: 1 = Reject True | 2 = Accept False

✓ 4. Level of Significance (α) & Confidence Level

Confidence Level = $1 - \alpha$

If $\alpha = 0.05$ [5%], Confidence = 95%. | If $\alpha = 0.01$ [1%], Confidence = 99%.

✓ 5. Z-test vs t-test - The Border Guard

Look only at **n** [sample size]. **The border is 30!**

- **$n \geq 30$** : Large sample → **Z-test**
- **$n < 30$** : Small sample → **t-test**

✓ 6. Tailed Tests & P-Value Rule

Arrows: $>$ or $<$ [One direction] = **One-tailed** | $\neq$ [Two directions] = **Two-tailed**.

P-Value vs α : **Small p = Problem $\rightarrow$ Reject H_0** [$p \leq \alpha$ means Reject H_0].

✓ 7. Estimator vs Estimate

Estimator: The Formula [e.g., $\bar{x}$] | **Estimate:** The Final Value [e.g., 52].

5-SECOND REVISION SHEET

- ✓ **H_0 :** Equality Sign [=]
- ✓ **Type I:** Reject True
- ✓ **Type II:** Accept False
- ✓ **Confidence:** $1 - \alpha$
- ✓ **Z-test:** $n \geq 30$
- ✓ **Reject H_0 :** $p\text{-value} \leq \alpha$

THUNDERSTUDY: KEYWORDS > CALCULATIONS